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#67 (1571345184): Interpretable Machine Learning for Child Stunting Assesment

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Interpretable Machine Learning for Child Stunting Assessment

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Abstract—Stunting among children under five remains a major public health challenge in Indonesia, demanding solutions that support both province-level surveillance and individual-level nutritional screening to guide targeted intervention. However, existing approaches typically address these needs in isolation, relying on isolated, single-purpose models that lack interpretability and fail to concurrently address regional nutritional vulnerability alongside individual anthropometric status. Here, we propose an interpretable, integrated machine learning framework that combines clustering, regression, and classification as complementary but independently trained components to comprehensively assess child stunting across spatial and individual scales. The framework draws on two distinct datasets: a provincial spatial-demographic dataset (34 provinces and 514 districts/cities) and an individual child anthropometric dataset, capturing regional nutritional indicators and individual growth measures, respectively. In the clustering task, we evaluated K-Means, Gaussian Mixture Models, and Hierarchical clustering using Elbow and Silhouette analysis, and identified Hierarchical clustering as the best-performing method, achieving a Silhouette score of 0.3199 and a Davies-Bouldin Index of 0.9877. Six regression algorithms were benchmarked to predict Height-for-Age Z-scores (HAZ) and Weight-for-Age Z-scores (WAZ), among which Random Forest and CatBoost achieved good performance. Nutritional status classification compared five

I. INTRODUCTION

Stunting among children under five remains a significant global health challenge, with 96.7% of the approximately 149 million cases recorded in 2020 occurring in low and middle income countries [1], [2]. In Indonesia, the 2024 Indonesian Nutritional Status Survey (SSGI) [3] reported a national stunting prevalence of 19.8%, close to the 20% threshold defined by the World Health Organization as indicative of a public health problem, while several provinces in eastern Indonesia still recorded prevalence rates exceeding 30%, a pattern already evident in the 2018 Basic Health Research (Riskesdas) data showing a national figure of 30.8% [4]. This persistent burden stands in contrast to the target set by the National Medium Term Development Plan (RPJMN) [5], which mandates a reduction of national stunting prevalence to below 14% by 2029. As an archipelagic country with substantial geographic, social, economic, and health related heterogeneity across regions, Indonesia faces the additional challenge that uniform, one size fits all intervention strategies often fail to address region specific determinants of child malnutrition, underscoring the need to understand both the spatial distribution and temporal trends of stunting risk across administrative areas.

case.

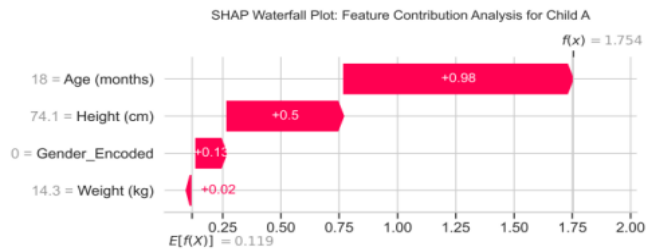


Fig. 8. SHAP waterfall plot illustrating individual feature contribution for a representative child case.

For this individual case, with an age of 18 months, a height of 74.1 cm, Gender_Encoded of 0, and a weight of 14.3 kg, Age contributed the largest positive impact on the predicted output (+0.98), followed by Height (+0.5), Gender_Encoded (+0.13), and Weight (+0.02). The cumulative contribution of these four features shifted the prediction from the baseline expected value of 0.119 to a final output of 1.754, illustrating how individual feature values combine to drive a single prediction, consistent with the global feature importance pattern observed in Figure 7, where Age and Height are the dominant contributors.

D. Dashboard Implementation

To translate the analytical results into a practical decision support tool, the trained clustering, regression, and classification models were deployed within an interactive web based dashboard (see Figure 9), consisting of three main interfaces as described in Section II.

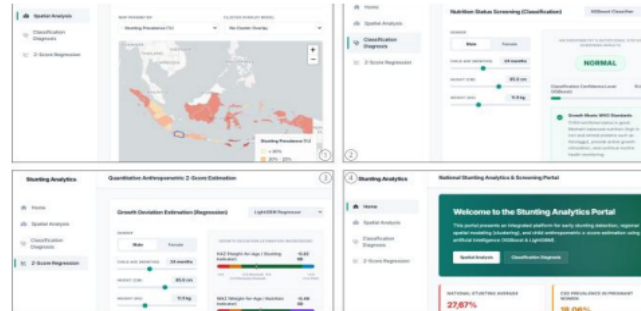


Fig. 9. Interface of the web dashboard. (1) Clusterin; (2) classification; (3) regression; (4) Home page.

IV. CONCLUSION

This study proposed an interpretable, integrated machine learning framework for child stunting assessment, combining clustering, regression, and classification within a single unified pipeline supported by two distinct datasets, the

and classification (99.98% accuracy) tasks, with CatBoost a close second throughout. What ties these results together is less the specific algorithm choice than the recurring dominance of height and age as predictive features, a signal reinforced independently by SHAP analysis on both the LightGBM regressor and XGBoost classifier, suggesting the underlying growth-status signal is robust across model families rather than an artifact of any single algorithm.

Several limitations should be acknowledged. The clustering task relies on a relatively small number of observations at the provincial level, which may explain the non-monotonic behavior observed in the Within-Cluster Sum of Squares curve and limits the extent to which the resulting risk clusters can be generalized beyond the observed period. The CAdataset, while large in size, is sourced from a single public dataset rather than a nationally representative individual level survey, which may limit the generalizability of the regression and classification results to the broader Indonesian child population. In addition, this study does not establish a direct quantitative link between the regional level clustering results and the individual level regression and classification results, as the two tasks rely on separate datasets.

Future work may address these limitations by incorporating a nationally representative individual level dataset, extending the temporal coverage of the regional level data to examine trends over time, and exploring methods to more directly connect regional vulnerability profiles with individual level nutritional outcomes within a single integrated model.

DATA AND CODE AVAILABILITY

The source code for the experiments conducted in this study is publicly available in a GitHub repository. The resulting interactive dashboard is also publicly accessible. Source code: github.com/haerulyudaaditiya/Stunting-Detection-ML-Spatial-Analysis. Dashboard: haerulyudaaditiya.github.io/Stunting-Detection-ML-Spatial-Analysis

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Purwarupa v1











Purwarupa v2







